Day 02

File System Navigation

```
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```

Figure 1: pwd (print working directory)
Displays the full path of the directory you are currently working in.

```
┌──(kali㉿kali)-[~]
└─$ ls -la
total 348
drwx------ 27 kali kali  4096 Nov 11 18:12 .
drwxr-xr-x  3 root root  4096 Apr 15  2024 ..
drwx------  7 kali kali  4096 Jan 12  2025 .BurpSuite
-rw-------  1 kali kali     0 Apr 15  2024 .ICEauthority
-rw-------  1 kali kali    49 Nov 11 18:12 .Xauthority
drwxrwxr-x 22 kali kali  4096 Dec 31  2024 .ZAP
-rw-------  1 kali kali   806 Sep 30  2024 .bash_history
-rw-r--r--  1 kali kali   220 Apr 15  2024 .bash_logout
-rw-r--r--  1 kali kali  5551 Apr 15  2024 .bashrc
-rw-r--r--  1 kali kali  3526 Apr 15  2024 .bashrc.original
drwxr-xr-x 16 kali kali  4096 Mar 18  2025 .cache
drwxr-xr-x 21 kali kali  4096 Nov 11 13:55 .config
drwx------  3 kali kali  4096 Nov  8  2024 .dbus
-rw-r--r--  1 kali kali    35 Apr 15  2024 .dmrc
-rw-------  1 root root 12288 Oct  5  2024 .docker-compose.yml.swo
-rw-r--r--  1 kali kali 11759 Apr 15  2024 .face
lrwxrwxrwx  1 kali kali     5 Apr 15  2024 .face.icon -> .face
drwx------  3 kali kali  4096 Apr 15  2024 .gnupg
drwx------  2 kali kali  4096 Nov  8  2024 .gvfs
drwxr-xr-x  4 kali kali  4096 Sep 30  2024 .java
-rw-------  1 kali kali    49 Nov 11 15:18 .lesshst
drwxr-xr-x  5 kali kali  4096 Nov  8  2024 .local
drwx------  4 kali kali  4096 Sep 18  2024 .mozilla
drwxr-xr-x 11 kali kali  4096 Oct 28  2024 .msf4
-rw-r--r--  1 kali kali   807 Apr 15  2024 .profile
drwxr-xr-x  4 kali kali  4096 Nov  4  2024 .recon-ng
-rw-r--r--  1 kali kali     0 May  8  2024 .sudo_as_admin_successful
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-clipboard-tty7-control.pid
-rw-r-----  1 kali kali     5 Nov 11 18:12 .vboxclient-clipboard-tty7-service.pid
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-display-svga-x11-tty7-control.pid
-rw-r-----  1 kali kali     5 Nov 11 18:12 .vboxclient-display-svga-x11-tty7-service.pid
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-draganddrop-tty7-control.pid
-rw-r-----  1 kali kali     5 Nov 11 18:12 .vboxclient-draganddrop-tty7-service.pid
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-hostversion-tty7-control.pid
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-seamless-tty7-control.pid
-rw-r-----  1 kali kali     5 Nov 11 18:12 .vboxclient-seamless-tty7-service.pid
-rw-r-----  1 kali kali     6 Nov 11 18:12 .vboxclient-vmsvga-session-tty7-control.pid
-rw-------  1 kali kali  1042 Jan 27  2025 .viminfo
-rw-------  1 kali kali  4897 Nov 11 18:12 .xsession-errors
-rw-------  1 kali kali  5967 Nov 11 15:43 .xsession-errors.old
drwxrwxr-x  2 kali kali  4096 Nov  8  2024 .zenmap
-rw-r--r--  1 kali kali   336 Nov  8  2024 .zprofile
-rw-------  1 kali kali   940 Nov 11 15:43 .zsh_history
-rw-rw-r--  1 kali kali     0 Nov 10 10:53 .zsh_history_old
-rw-rw-r--  1 kali kali 12245 Nov 10 10:52 .zshr_history_backup
```

Figure 2: ls -la
Lists all files and directories in the current directory,including hidden files.



Figure 3: cd /etc

Changed your current directory to /etc. Allows you to navigate to different directories and inspect system configuration files.

File Operations



Figure 4: touch test.txt

Creates an empty file named “test.txt”. If the file already exists, it updates its modification timestamp.

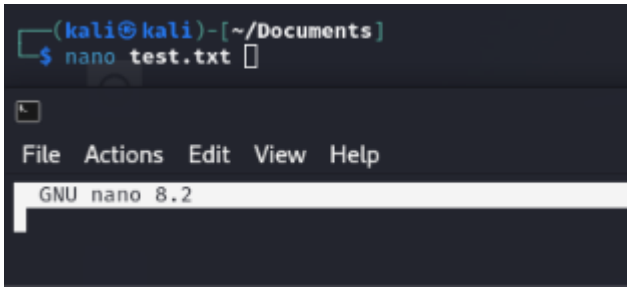


Figure 5: nano test.txt

Opens “test.txt” in the nano text editor. You can write, edit, save text inside the file.

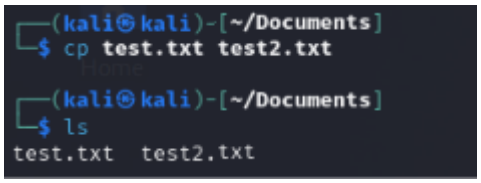


Figure 6: cp test.txt test2.txt

Copies “test.txt” into a new file named “test2.txt”. Creates a duplicate while keeping the original file.



Figure 7: mv test2.txt /tmp
Moves “test2.txt” into the /tmp directory. Moves or renames files. “/tmp” is commonly used for temporary files.



Figure 8: rm test.txt
Deletes the file named “test.txt”

User & Permissions



Figure 9: whoami
Displays the username of the current user.

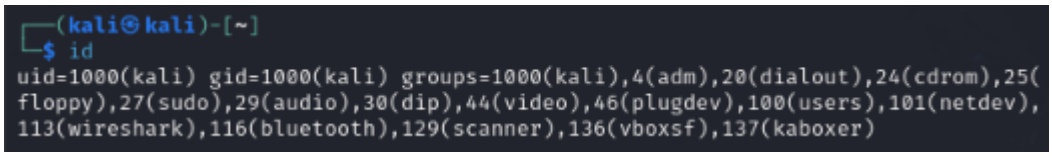


Figure 10: id
Displays the current user’s UID (User ID), GID (Group ID) and group membership. Helps understand the user’s identity and permissions within Linux.

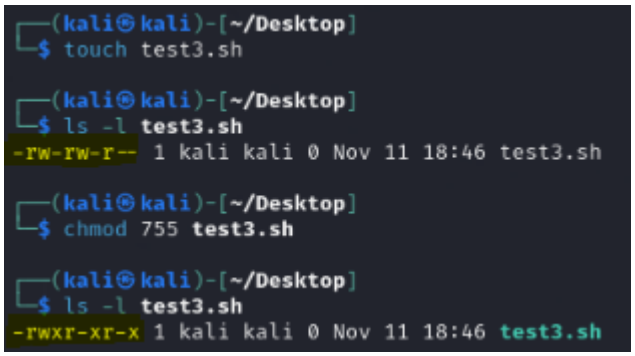


Figure 11: chmod 755 file
Changes the file permission of a file, allowing the owner to read,write and execute.
7 – Owner can read, write and execute.
5 – Group can read and execute.
5 – Others can read and execute.

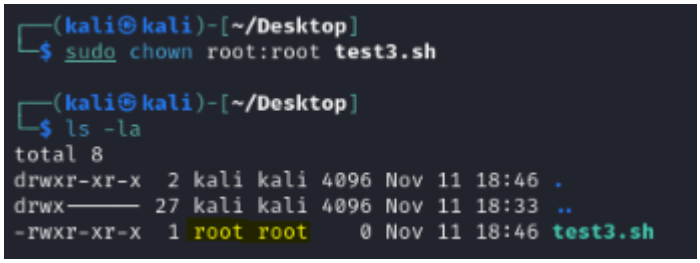


Figure 12: chown root:root file

Changes the ownership of a file to the root user and root group.

Processes

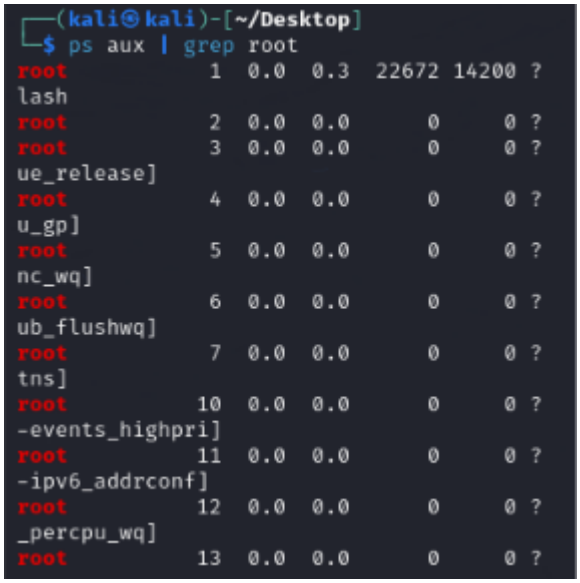


Figure 13: ps aux | grep root

Displays running processes and filters the output to show lines containing the word “root”.

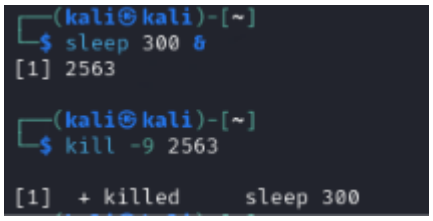


Figure 14: kill -9 <pid>

Forcefully terminates a process using its process ID.

Hands on Task

```
┌──(kali㉿kali)-[~/Desktop]
└─$ mkdir Security_Project

┌──(kali㉿kali)-[~/Desktop]
└─$ cd Security_Project

┌──(kali㉿kali)-[~/Desktop/Security_Project]
└─$ mkdir Reports Logs Scripts Evidence

┌──(kali㉿kali)-[~/Desktop/Security_Project]
└─$ ls -la
total 24
drwxrwxr-x 6 kali kali 4096 Nov 11 20:09 .
drwxr-xr-x 3 kali kali 4096 Nov 11 20:08 ..
drwxrwxr-x 2 kali kali 4096 Nov 11 20:09 Evidence
drwxrwxr-x 2 kali kali 4096 Nov 11 20:09 Logs
drwxrwxr-x 2 kali kali 4096 Nov 11 20:09 Reports
drwxrwxr-x 2 kali kali 4096 Nov 11 20:09 Scripts

┌──(kali㉿kali)-[~/Desktop/Security_Project]
└─$ tree
.
├── Evidence
├── Logs
├── Reports
└── Scripts
```

Figure 15: Create a directory structure for a “Security Project”

```
┌──(root㉿kali)-[/home/kali]
└─# adduser analyst
info: Adding user `analyst' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `analyst' (1001) ...
info: Adding new user `analyst' (1001) with group `analyst (1001)' ...
info: Creating home directory `/home/analyst' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for analyst
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `analyst' to supplemental / extra groups `users'
info: Adding user `analyst' to group `users' ...

┌──(root㉿kali)-[/home/kali]
└─# id analyst
uid=1001(analyst) gid=1001(analyst) groups=1001(analyst),100(users)
```

Figure 16: Create a non root user.

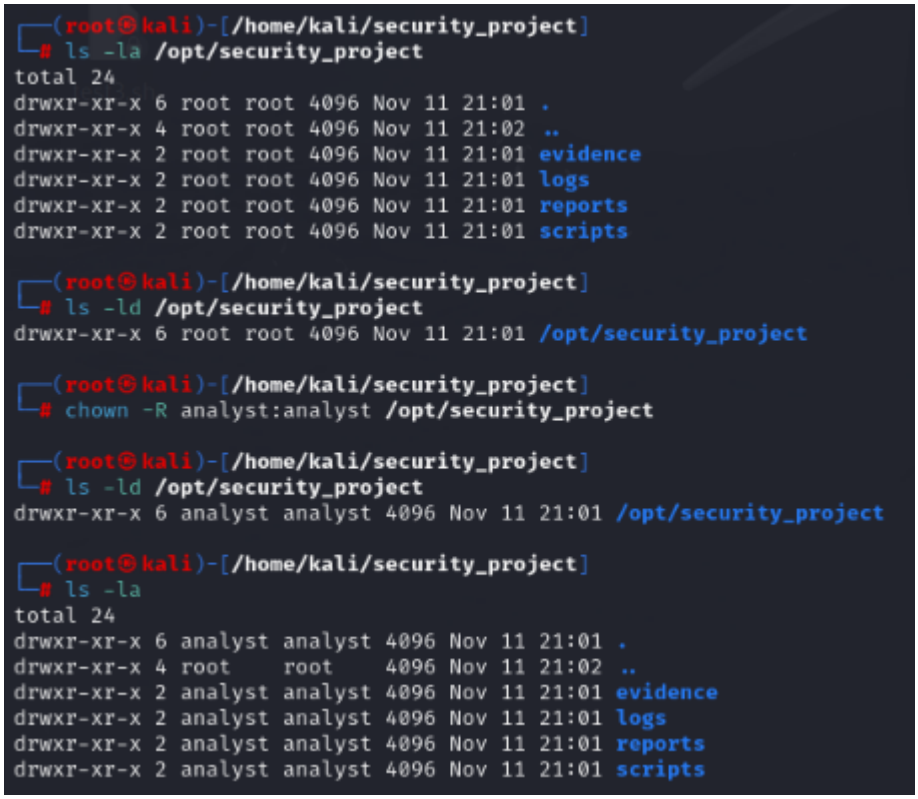


Figure 17: Created a non root user (analyst), changed the ownership of an existing folder from “root” to “analyst” using chown.

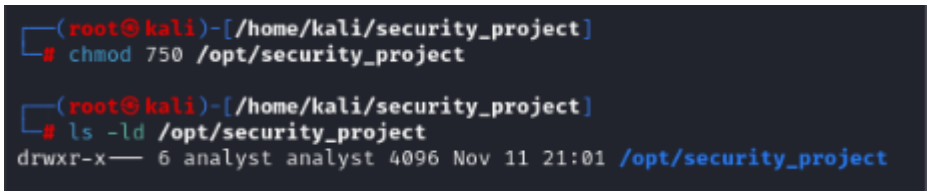


Figure 18: Assigned permissions to user using chmod 750.